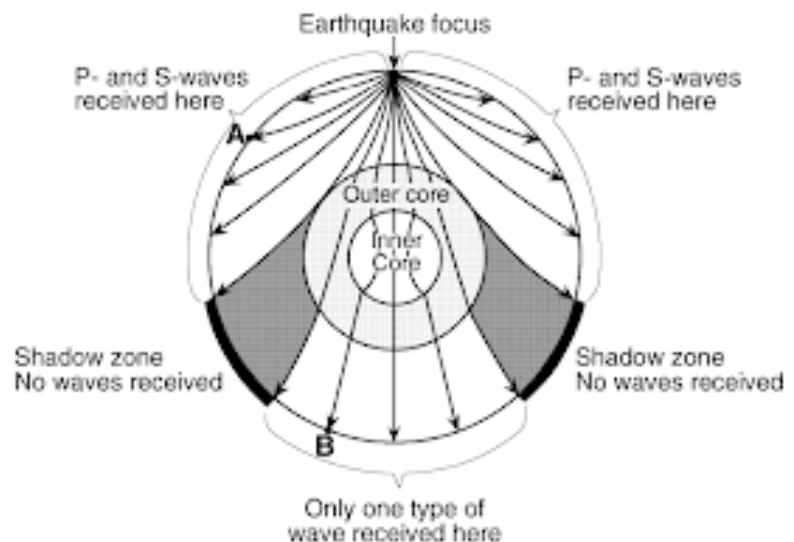
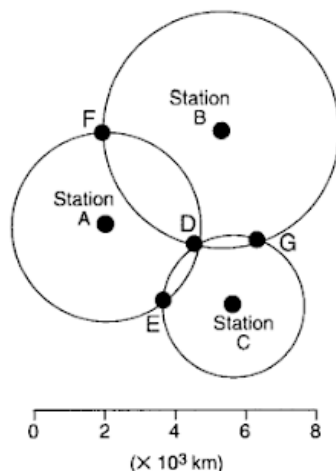




### Earth Science Unit 2 Study Guide

1. What evidence supports the theory of continental drift? same rocks & fossils on different continents, continents fit together, evidence of glaciers in now tropical areas (and evidence of past tropical climates near the poles)
2. List the layers of the Earth in order starting at the surface, including minor layers. inner core, outer core, lower mantle, asthenosphere (upper mantle), lithosphere, crust
3. What two layers make up the lithosphere? crust + very top layer of mantle
4. How is the asthenosphere different from the lithosphere? it is plastic (very slowly moving solid)
5. List the main layers of the Earth in order from thinnest to thickest. crust, inner core, outer core, mantle
6. Describe the force that causes plate tectonics to move and shift. convection currents in the asthenosphere
7. What happens at an oceanic–oceanic convergent boundary, and what features does this form? plates collide, subduction, trenches, island arc volcanoes
8. What forms when two continental plates collide? mountains
9. What happens at a divergent boundary, and what features does this form? plates move apart, magma rises up forming new crust at mid-ocean ridge
10. The San Andres Fault in southern California is an example of what type of plate boundary? transform
11. Why do oceanic plates subduct beneath continental plates? the basaltic rocks are more dense

12. At what type of plate boundary is the youngest ocean floor rock found, and why? **divergent, magma rises at mid ocean ridges making new crust**
13. Name the mountain range formed by the collision of the Indian and Eurasian plates. **Himalayas**
14. Describe seafloor spreading. **plates move apart at mid-ocean ridges pushing the plates apart making the ocean larger (see the mid-atlantic ridge)**
15. At what type of plate boundary does seafloor spreading occur? **oceanic-oceanic divergent**
16. What is the epicenter of an earthquake, and how is the focus different? **focus is where the fault actually moves, epicenter is the point on the surface directly above that**
17. How many seismographs are needed to locate an epicenter, and why? **3 - you need 3 (or more) so they can all meet at one point**
18. Which seismic waves are the fastest to arrive? **primary**
19. Base your answer to the following question on the diagram below, and on the left, which represents seismic stations A, B, and C. Which letter represents the epicenter of this earthquake? **D**



20. Using the diagram above and to the right, why can P waves travel through Earth's core, but S waves cannot? S-waves stop at the liquid outer core

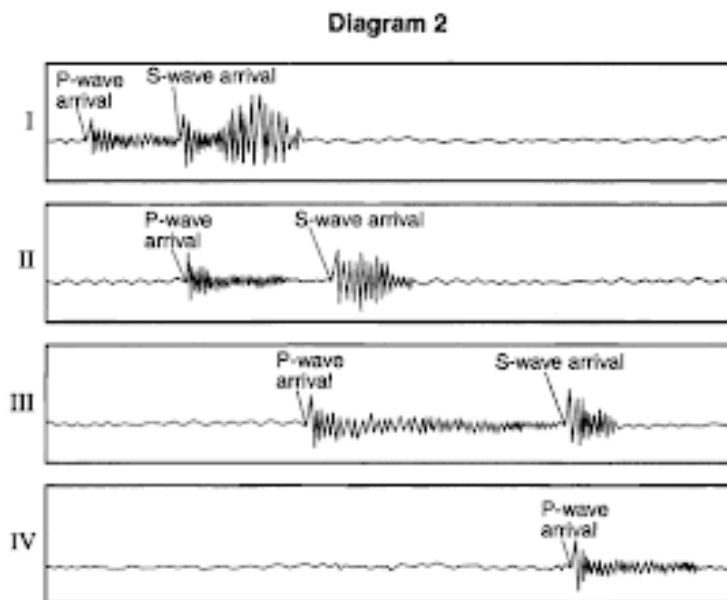
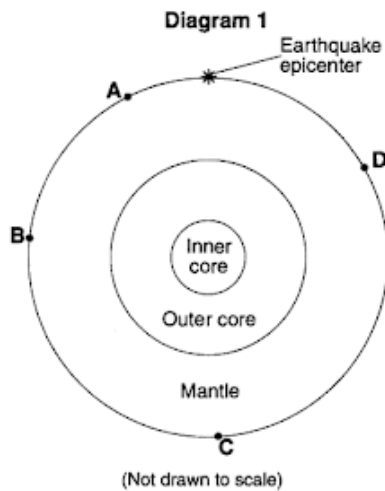
21. How does the distance from the epicenter affect the time between arrival of P and S waves? the further away, the longer the difference between the arrival of the P and S waves

22. Why are earthquakes felt more strongly at the epicenter than farther away? that's the place directly above where the earthquake happens so closest to where the energy was released

23. Which type of earthquake causes the most severe damage: shallow or deep focus? Why? shallow, closer to the surface will cause more shaking (assuming the energy is the same as a deeper quake)

24. Why do most earthquakes occur along plate boundaries? that is where the motion occurs causing rocks to slide past each other

*Base your answers to the next few questions on the diagrams below. Diagram 1 represents a cross section of Earth and its interior layers. The asterisk (\*) shows the location of an earthquake epicenter. Letters A through D are seismic stations on Earth's surface. Diagram 2 shows four seismograms labeled I, II, III, and IV, which were recorded at seismic stations A, B, C, and D during the same time interval.*



25. Create a list matching the seismograms with the seismic stations where they were recorded. **A-I, B-III, C-IV, D-III**

*For the next few earthquake questions, use the earthquake P-wave and S-wave graph on the last page of this file.*

26. Station D is 8000 kilometers from the earthquake epicenter. How long did it take for the first P-wave to travel from the epicenter to station D? **11 min 20 sec**

27. The first S-wave arrived at a seismograph station 11 minutes after an earthquake occurred. How long after the arrival of the first P-wave did this first S-wave arrive? (about) 4 minutes 20 sec

28. An earthquake's first P-wave arrives at a seismic station at 12:00:00. This P-wave has traveled 6000 kilometers from the epicenter. At what time will the first S-wave from the same earthquake arrive at the seismic station?

(about) 12:07:30

29. How long would it take for the first S-wave to arrive at a seismic station 4,000 kilometers away from the epicenter of an earthquake? 12 minutes 40 sec

30. A seismic station 4,000 kilometers from the epicenter of an earthquake records the arrival time of the first P-wave at 10:00:00. At what time did the first S-wave arrive at this station? about 10:05:40

For each fault, note the movement of the hanging wall and foot wall and the stress that causes each):

31. a normal fault compression, hanging wall moves up

32. a reverse fault tension, hanging wall moves down

33. a strike-slip fault shearing, rocks on either side of the fault move past each other

34. Why does magma from Earth's interior rise toward the surface? less dense than solid rocks

35. What is a hotspot? pool of magma under the plates that cause volcanoes (think: Hawaii)

36. Where are hotspots located relative to plate boundaries? randomly - away from the edges

37. What feature does a hotspot form on the Earth's surface? shield volcanoes (Hawaii)

38. How does the age of the Hawaiian Islands relate to the distance from the hot spot (that is currently under the big island of Hawaii) the youngest is the big island, they get older as you move away from it